

Applying Brauer Data Analysis to Artin's Braid Group with Applications in Quantum Entanglement and Quantum Computing

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Abstract. Brauer Data Analysis (BDA) is a novel data analysis technique. It applies algebraic and combinatorial invariants of so-called Brauer configuration algebras for data analysis. This talk explains how elements of Artin's Braid group can be interpreted as specialized computers for constructing such algebras, and how this setting allows us to interpret quantum circuits and certain quantum entanglement states as indecomposable projective modules associated with some Brauer configuration algebras. Applications in the CAPTCHA industry are also described.

Keywords. BDA, Braid group, Borromean rings, CAPTCHA, Emerging image, Quantum entanglement, Quantum computing, Topological quantum computing.

Brauer configuration algebras were introduced in 2017 by Green and Schroll [7] to investigate algebras of wild and tame representation type. The analysis of their algebraic invariants such as their dimension, the dimension of their centers, and the structure of their indecomposable projective modules is said to be a Brauer analysis of the data (BDA) induced by them. Such an analysis allows for finding connections between the representation theory of algebras, integer partitions theory, graph energy theory, branched covering theory [3], snake graphs [6], and the Yang-Baxter equation theory [5], with practical applications in

- Manufacturing textile, freight industry [9].
- Cryptography [1].
- Quantum Computing [2].
- Cybersecurity [1].
- Music Composition [4].

This talk explains how elements of Artin's braid group (braids) can be viewed as specialized computers for constructing appropriate Brauer configuration algebras. Actually, Kauffman and Lomonaco [8] have interpreted generalizations of Artin's Braid group as quantum gates giving rise to the so-called topological quantum computing. In this talk, the family of Borromean-Brauer configuration algebras is described, applying to them a corresponding BDA by adding new tools, such as the algebraic and topological entropies, and associated covering multigraphs. We also interpret the Borromean knot as a GHZ (Greenberger-Horne-Zeilinger) quantum state and show how this machinery can construct emerging images which are associated with some open problems in AI and some based-reasoning CAPTCHAs.

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